

① Design a Class Hierarchy (Pseudocode)

pseudocode

START

CLASS person

DATA MEMBERS:

name
age

END CLASS

CLASS Student INHERITS person

DATA MEMBERS:

register Number
marks [5]
average

METHOD CalculateAverage()

Sum $\leftarrow$ 0

FOR i $\leftarrow$ 0 TO 4

Sum $\leftarrow$ Sum + marks[i]

END FOR

average $\leftarrow$ Sum/5

END METHOD

END CLASS

CLASS ResearchStudent INHERITS Student

DATA MEMBERS:

researchArea

METHOD DisplayCompleteDetails()

DISPLAY "Name :", name

DISPLAY "Age :", age

DISPLAY "Register Number :", registerNumber


```

        DISPLAY "Research Area :", researchArea
        DISPLAY "Average Marks :", average
    END METHOD
END CLASS

```

MAIN PROGRAM

CREATE object rs OF ResearchStudent

INPUT rs.name

INPUT rs.age

INPUT rs.registerNumber

FOR i ← 0 TO 4

 INPUT rs.marks[i]

END FOR

INPUT rs.researchArea

CALL rs.CalculateAverage()

CALL rs.DisplayCompleteDetails()

STOP

Input:

Enter Name: Madhur;

Enter Age : 20

Enter Register Number: 1924725

Enter Marks in 5 Subjects:
85, 90, 88, 92, 95

Enter Research Area: Artificial
Intelligence

Output:

Name: Madhur;

Age : 20

Register Number: 1924725

Research Area: Artificial
Intelligence

Average Marks: 90

(Q2) Complete the Missing pseudocode

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$ THEN

FOR $k \leftarrow j$ TO $N-2$

$A[k] \leftarrow A[k+1]$

ENDFOR

$N \leftarrow N-1$

$j \leftarrow j-1$

ENDIF

ENDFOR

ENDFOR

PRINT Modified Array

END

Input :

$N = 6$

$A = [2, 4, 2, 5, 4, 6]$

Output :

Modified Array : 2, 4, 5, 6

(Q3) Debug the pseudocode

Corrected pseudocode

CLASS Area

METHOD area(length)

RETURN length * length

END METHOD

METHOD area(length, breadth)

RETURN length * breadth

END METHOD

END CLASS

Incorrect line:

RETURN length * breadth

Correct line:

RETURN length * length

In METHOD area(length), the variable breadth is undefined. Since it calculates the area of a square, it should return length x length.

(Q4)

Trace the Execution

Corrected Pseudocode

BEGIN

Sum $\leftarrow$ 0

FOR $i \leftarrow 1$ to 5

IF $i \bmod 2 == 0$ THEN

Sum $\leftarrow$ Sum + i ;

ELSE

Sum $\leftarrow$ Sum - i ;

ENDIF

ENDFOR

PRINT Sum

END.

Trace Table

i	Even/odd	operation	Sum
1	odd	$0 - 1$	-1
2	even	$-1 + 2$	1
3	odd	$1 - 3$	-2
4	Even	$-2 + 4$	2
5	odd	$2 - 5$	-3

Output : -3

(Q5) Write Efficient Efficient pseudoCode

pseudoCode

BEGIN

READ N

READ ARRAY A

largest $\leftarrow -\infty$

SecondLargest $\leftarrow -\infty$

FOR $i \leftarrow 0$ TO $N-1$

IF $A[i] > \text{largest}$ THEN

SecondLargest $\leftarrow \text{largest}$

largest $\leftarrow A[i]$

ELSE IF $A[i] > \text{SecondLargest}$ AND $A[i] \neq$

largest THEN

SecondLargest $\leftarrow A[i]$

ENDIF

ENDFOR

PRINT SecondLargest

END

Input:

$N = 6$

$A = [10, 25, 18, 25, 15, 20]$

output:

20

(Q6)

Convert Problem Statement into Pseudocode

BEGIN

INTERFACE Payment

METHOD pay(amount)

END INTERFACE

CLASS CreditCard Implements Payment

METHOD pay(amount)

DISPLAY "payment of", amount, "done using
Credit Card"

END METHOD

END CLASS

CLASS UPI IMPLEMENTS Payment

METHOD pay(amount)

DISPLAY "payment of", amount, "done using
UPI"

END METHOD

END CLASS

MAIN

READ choice

READ amount

IF choice = 1 THEN

payment ← CreditCard

ELSE IF choice = 2 THEN

payment ← UPI

ELSE

payment ← NetBanking

ENDIF

CALL payment.pay(amount)

END

Input :

Enter choice :

1. Credit Card

2. UPI

3. Net Banking

Choice : 2

Amount : 50000

Output :

payment- of 50000 done using UPI

(Q7) Complete the pseudocode

```
CLASS Animal
    METHOD speak()
        PRINT "Animal"
    END METHOD
END CLASS
```

```
CLASS Dog INHERITS Animal
    METHOD speak()
        PRINT "Barks"
    END METHOD
END CLASS
```

```
Animal obj ← Dog()
```

```
obj.speak()
```

output

Barks

(Q8)

Find Logical Errors

Logical Mistakes:

- The constructor should accept id and name as parameters.
- Use this.id and this.name to assign the parameters values to the object's data members.

Correct Constructor:

CONSTRUCTOR(id, name)

 this.id ← id

 this.name ← name

END CONSTRUCTOR

(Q9) Case Study (Integrated problem)

BEGIN

CLASS Vehicle

DATA:

VehicleNO

ownerName

METHOD display()

DISPLAY VehicleNO

DISPLAY ownerName

END METHOD

END CLASS

CLASS Car INHERITS Vehicle

DATA:

FuelType

METHOD display()

DISPLAY VehicleNO

DISPLAY ownerName

DISPLAY fuelType

END METHOD

END CLASS

CLASS Truck INHERITS Vehicle

DATA:

loadCapacity

METHOD display()

DISPLAY VehicleNO

DISPLAY ownerName

DISPLAY loadCapacity

END METHOD

END CLASS


```

MAIN
READ N
CREATE ARRAY VEHICLES[N]
FOR i ← 0 TO N-1
    READ type
    IF type = "Car" THEN
        CREATE TRUCK t
        READ t.vehicleNo
        READ t.ownerName
        READ t.loadCapacity
        VEHICLES[i] ← t
    ENDIF
ENDFOR
FOR i ← 0 TO N-1
    CALL VEHICLES[i].display()
ENDFOR
END

```

Input

N = 2

Car

TN01AB1234

Ravi

petrol

Truck

TN02CD5678

kumar

15 Tons

Output

Vehicle No : TN01AB1234

owner Name : Ravi

Fuel Type : petrol

Vehicle No : TN02CD5678

owner Name : kumar

Load Capacity : 15 Tons